

Arjun Thakur

Principal / Lead AI Engineer | Architect & Fractional CTO | Agentic AI Developer & Corporate AI Trainer | Founder | ex-Amazon, ex-Agoda

Available for: Lead / Principal AI Engineer | Architect | Fractional CTO | Forward-Deployed AI Engineer | Corporate AI training, workshops & enablement | Full-time & freelance | Remote (global) | Based in India, open to international travel & relocation

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SUMMARY

- AI engineer, principal-level architect, and corporate AI trainer who both builds production agentic systems and trains enterprise engineering teams to do the same — LangGraph agent graphs, multi-agent orchestration, RAG pipelines, and AI automation, delivered hands-on to some of the world's largest technology and manufacturing organizations.
- Designs and delivers advanced Agentic AI & AI-automation programs — concept-first teaching, live coding, hands-on labs, and production-ready capstones — grounded in 15+ years of real shipping experience, not slideware.
- Runs two distinct programme tracks so an entire organization can be upskilled, not only its developers — a code-first Agentic AI & RAG engineering track for software engineers, and a no-code / low-code AI automation track for non-technical business teams.
- 15+ years building production systems at Amazon, Agoda, and high-growth startups across India, Southeast Asia, and the US; brings senior-engineering rigor to AI — distributed systems, scale, security, and cost engineering.
- Runs LangSmith evals, retrieval-precision metrics, and agent traces before calling anything production-ready — engineering discipline, not vibes-based prompting.

CLIENTS & TRAINING DELIVERY

LG Electronics • LG Software (engagement in pipeline) • Simplilearn • PhysicsWallah • Masai School

RECENT AI TRAINING & ENABLEMENT

- Ran a 4-day LLM / SLM engineering deep-dive for 30 senior engineers at LG Electronics — Transformers & self-attention, RAG, LangGraph dialogue management, and inference optimization (quantization, KV-cache), taught concept-first with live code and supervised parameter tuning.
- Delivered a workshop briefing on Agentic AI and the Industry 4.0 → 5.0 shift for LG Electronics, Noida — AI agents vs. automation, real-world industrial agent use cases, and human-machine collaboration for smart manufacturing.
- Design and deliver hands-on Agentic AI & AI-automation curricula — agent design, multi-agent orchestration, tool/function calling, MCP integration, RAG, evaluation & observability, and governance — with each engineer shipping a working agentic prototype.

TRAINING PROGRAMS

- Code-first Agentic AI & RAG engineering — for working software engineers. Retrieval-Augmented Generation, AI agents built from first principles, LangGraph / LangChain or AWS Bedrock, MCP, LLM evaluations, guardrails, observability and cost control. Typically ~75% hands-on with a production-shaped capstone.
- No-code / low-code AI automation — for non-technical teams across operations, finance, HR, support and quality. AI agents, RAG document search, multi-agent orchestration, triggers and approvals, error handling and governance. No programming required.
- Every program is scoped to the client: duration, depth, stack and worked examples are set after a scoping call, and run on the client's own domain, documents and systems. Delivered on-site or live online, worldwide.

CORE SKILLS

- **Agentic AI & LLMs:** LangChain, LangGraph (stateful agent graphs), Microsoft AutoGen, multi-agent orchestration, RAG, semantic FAQ caching, function/tool calling, MCP (Model Context Protocol), A2A patterns, prompt engineering, evals, agent observability (LangSmith), guardrails & verifier patterns
- **Training & Enablement:** Corporate AI training & workshops, curriculum design, hands-on labs & capstones, technical upskilling programs, LLM/SLM & Agentic AI bootcamps, executive briefings, conference speaking (IITs, NITs, industry events), no-code / low-code AI enablement (n8n, NotebookLM) for non-technical teams
- **LLM Providers & Voice:** OpenAI (GPT-4o / 4o-mini, Realtime API, Embeddings), Azure OpenAI, Anthropic Claude, Gemini, Whisper, WebRTC voice agents, streaming + interruption handling
- **Vector / Retrieval:** pgvector, embeddings (text-embedding-3, Cohere, BGE), hybrid search, chunking strategies, cosine-similarity caching, retrieval evals

- **AI-Native Backend:** Python 3.11+, FastAPI (async), Pydantic v2, asyncpg, pytest-asyncio, Java 21, Spring Boot, Spring Security, Hibernate JPA, Scala
- **Architecture:** Microservices, distributed systems, event-driven (Kafka), HLD/LLD, design patterns, SOLID, API design (REST + OpenAPI), TDD
- **Data:** PostgreSQL, pgvector, Cassandra, DynamoDB, Redis, MongoDB, SQL tuning, PgBouncer, Supabase (Auth + RLS)
- **Cloud & DevOps:** AWS Certified Solutions Architect — S3, ECS, Lambda, IAM, Serverless | Docker, Kubernetes, GitHub Actions, Jenkins, Railway, Vercel, Grafana, ELK
- **AI-Assisted Development:** Claude Code, Cursor, GitHub Copilot, v0.dev — shipping with AI as a daily collaborator
- **Leadership:** Strategic planning, hiring & team growth, mentoring, stakeholder management, agile/scrum, cross-functional delivery, founder-mode product + GTM

EXPERIENCE

Founder & AI Engineer | Yuvan (Voice-first AI Math Tutor)

Apr 2025 – Present | Remote • *Agentic AI | LangGraph | RAG | OpenAI Realtime | FastAPI | Next.js | Supabase | pgvector*

- Built and shipped Yuvan, a voice-first AI tutor for CBSE Class 10 students — onboarded 5 paying schools as design partners on a B2B2C model; product live in MVP with paid pilots.
- Designed the full agentic architecture: a LangGraph state machine per doubt — embed → semantic FAQ-cache lookup → RAG retrieval over NCERT chunks → GPT-4o-mini generation → math verifier → topic auto-tagger → persistence — with an explicit re-explain loop on detected student confusion.
- Engineered a WebRTC voice pipeline on OpenAI Realtime (STT + TTS + VAD + barge-in) with backend-minted ephemeral tokens; achieved sub-1.5s p95 first-audio latency while keeping all reasoning, RAG, and verification server-side.
- Built a RAG layer on Postgres + pgvector (Supabase) and a global semantic FAQ cache (cosine ≥ 0.92) targeting ~30% hit rate to push per-session OpenAI cost below the unit-economics ceiling.
- Implemented a math-verifier safety net re-checking every numerical claim before the AI speaks it ($\leq 1s$ overhead) — the difference between a demo and a tutor schools trust.
- Owned everything end-to-end: architecture, code, infra (Railway + Vercel + Supabase), LangSmith tracing, evals, school sales, pricing, and pilot success.

Director, Full-Stack Engineering | PhysicsWallah

Nov 2024 – Apr 2025 | Bengaluru • *Java | Spring | Microservices | Python services | AWS | AI dev tooling*

- Hired and led a team of 15 engineers to build an in-house LMS using Java, Spring, Hibernate, TypeScript, microservices and AWS.
- Stood up CI/CD on Jenkins + Docker, lifting deployment frequency by 15% and cutting release-night fire-drills.
- Introduced Python/FastAPI services for ML-adjacent workloads and rolled out AI-assisted development (Copilot, code-review automation) across the team.

Independent AI Engineer & Builder | Agentic AI R&D

Nov 2023 – Nov 2024 | Remote • *LangChain multi-agent patterns | voice pipeline prototyping | embeddings benchmarking*

- Pivoted full-time into AI engineering and shipped an early AI math-teacher prototype to 900+ users — the validation work that became Yuvan.
- Benchmarked LangChain multi-agent patterns (router, supervisor, plan-and-execute) on real student conversations to pick Yuvan's architecture.
- Prototyped four voice-agent stacks (OpenAI Realtime, Whisper + ElevenLabs, Whisper + Coqui, Gemini Live) and ran latency/cost shootouts.
- Cost-engineered embeddings across OpenAI text-embedding-3, Cohere v3, BGE for retrieval precision vs. INR-per-1M-tokens.

Senior Engineering Manager (Contract) | ForaySoft (client: Wayfair)

Mar 2023 – Oct 2023 | Remote • *Java | Spring Security | Hibernate | PostgreSQL | Kubernetes | AWS*

- Led 10 engineers on a fixed-term migration of Wayfair's seller-promotions microservice, cutting system failures by 28%.
- Re-architected for scale on Kubernetes, supporting a 25% increase in transaction volume during peak promotions.

Vice President, Engineering (Hands-on) | Nolan EduTech

Oct 2021 – Nov 2022 | Remote • *Java | Spring | JPA | Microservices | AWS | TypeScript | Curriculum + GTM*

- Launched a coding-bootcamp vertical from zero to INR 30 crore (~\$3.6M) revenue in one year — owned curriculum, platform, and live teaching.
- Designed backend + full-stack programs aligned to industry hiring; ~95% placement across cohorts in the thousands.
- Taught live online batches on Java, backend systems, system design, and interview prep to thousands of learners.

Lead Developer | CoinSwitch Kuber

Dec 2020 – Oct 2021 | Remote • Python | FastAPI | Django | PostgreSQL | Redis | AWS | Microservices

- Migrated a monolithic crypto-trading app to microservices on Python, FastAPI, and Django on AWS — the same async stack behind current AI services.
- Rolled out Redis caching (~30% latency) and tuned PostgreSQL with PgBouncer pooling (~40% further); held the system stable through a 600K → 10M user surge.

Software Development Engineer II | Amazon

Jul 2019 – Oct 2020 | Chennai • Java | DynamoDB | AWS | Payments / Digital Goods

- Hardened payment security against CSRF attacks, reducing impact by 19.7%.
- Architected a throttling system on DynamoDB + Java that cut coupon-code misuse by 15.3% in month one.
- Integrated the Amazon Games payment flow into the Digital Payments Platform; improved payment-page accessibility across geographies and devices.

Earlier Experience

2015 – 2019 • Porch.com, Domino, Agoda, Hadoolytics

- Agoda (Sr. Software Developer, Bangkok): built an in-house A/B testing pipeline on Scala + Cassandra + Kafka processing millions of messages, cutting time-to-insight 22%.
- Porch.com (Contract SDE, Seattle): built a Facebook Marketplace integration from scratch — doubled service-quote requests, lifted revenue 13%.
- Hadoolytics: architected a real-estate platform from scratch on Java, Spring, AWS, Cassandra.

EDUCATION

- M.Tech, Computer Science & Engineering | NIT Jalandhar — CGPA 7.5/10. Taught and mentored thousands of undergraduates as a teaching assistant / instructor — the first chapter of a teaching thread that later became Nolan EduTech and PhysicsWallah.
- B.E., Computer Science & Engineering

CERTIFICATIONS

- AWS Certified Solutions Architect
- Advanced Architecting on AWS

SELECTED PROJECT HIGHLIGHTS

- **Yuvan** — voice-first AI tutor; LangGraph agent | RAG on pgvector | OpenAI Realtime WebRTC | math verifier | semantic FAQ cache | multi-tenant Supabase RLS | 5 paying schools.
- **Flash-sale system** — production-grade microservices simulating real concurrency, queueing, and back-pressure on Java 20 + React + TypeScript.
- **Online assessment platform** — HackerRank-style code-execution and grading on Java + Spring Boot + AWS, scaled for thousands of concurrent test-takers.